LEI LYU

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Education

University of Southern California, Viterbi School of Engineering — Los Angeles, CA
Ph.D. in Electrical Engineering; Advisor: Prof. Urbashi Mitra; GPA: 3.97/4.0 — *Expected May 2028*

Columbia University, Columbia Engineering — New York, NY
M.S. in Electrical Engineering; GPA: 4.198/4.3 — *May 2022*

Shanghai Jiao Tong University, UM–SJTU Joint Institute — Shanghai, China
B.S. in Electrical and Computer Engineering; GPA: 3.76/4.0 — *Aug. 2020*

Research Interests: Signal Processing, Wireless Communications, Physical-Layer Privacy, Machine Learning and Algorithm Design

Research Experience

University of Southern California, Viterbi School of Engineering — Los Angeles, CA
Research Assistant; Advisor: Prof. Urbashi Mitra — *Sep. 2024–Present*

Project: Hybrid Sparse–Diffuse Channel Estimation

- Developed a frequency-domain hybrid channel model that jointly represents specular multipath components and diffuse scattering in indoor wireless channels.
- Designed a hybrid atomic-norm and least-squares estimation algorithm to jointly recover sparse path parameters and diffuse channel components.
- Derived statistical properties of the proposed algorithms and Cramér–Rao bounds for channel response and parameter estimation.
- Evaluated the method using synthetic channel realizations and measured indoor channel data, with comparisons against conventional sparse-only channel estimation baselines.
- Produced two published conference papers and a journal manuscript under review from this research.

Project: Physical-Layer Privacy for Wireless Localization

- Developed a channel-state-information-free angle-spoofing strategy to protect a transmitter's location against eavesdroppers employing differential localization techniques.
- Designed a secret precoder matrix that manipulates angles of departure information observed by eavesdroppers while enabling an authorized receiver to reconstruct the original channel.
- Analyzed the asymptotic behavior of localization error as a function of the precoder design parameters.
- Evaluated the privacy–communication trade-off through theoretical analysis and simulations.
- Produced a first-author manuscript currently under review at *IEEE Signal Processing Letters*.

Shanghai Jiao Tong University, Emerging Computing Technology Laboratory — Shanghai, China
Research Assistant — *Apr. 2021–Sep. 2021*

Project: Logic Rewriting and Parallel Synthesis

- Analyzed logic-rewriting algorithms in the ABC Library and conducted experiments on benchmark circuits in C.
- Reviewed state-of-the-art logic-rewriting algorithms to understand their computational behavior and identify potential bottlenecks.
- Explored possible parallelization strategies for logic rewriting and preliminary ideas for a CUDA-based implementation.

Publications

Manuscripts Under Review

1. **L. Lyu**, M. Ferreira Da Costa, and U. Mitra, "Hybrid Sparse–Diffuse Channel Estimation and Cramér–Rao Bounds Analysis," submitted to *IEEE Transactions on Signal Processing*, May 2026.
2. **L. Lyu** and U. Mitra, "Channel State Information-Free Angle Spoofing for Differential Localization Privacy," submitted to *IEEE Signal Processing Letters*, Aug. 2026.

Published Conference Papers

1. **L. Lyu**, M. Ferreira Da Costa, and U. Mitra, "Cramér–Rao Bounds on Sparse–Diffuse Channel Estimation," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, May 2026.
2. **L. Lyu** and U. Mitra, "Hybrid Atomic Norm Sparse/Diffuse Channel Estimation," in *Proc. Allerton Conference on Communication, Control, and Computing*, Sep. 2025.

Work in Preparation

1. J. Gersey, J. Chae, **L. Lyu**, J. Zhang, U. Mitra, and P. Zhang, "RECOVER: Matrix-Based Reconstruction of Missing Vibration Signals for Robust Activity Recognition," in preparation.

Teaching Experience

Columbia University New York, NY

Teaching Assistant, Analysis of Algorithms, Department of Computer Science *Jan. 2022–May 2022*

- Supported Prof. Eleni Drinea in teaching Analysis of Algorithms, including grading assignments and coordinating course administration.
- Held office hours and explained algorithm design, correctness proofs, and complexity analysis to students.
- Collaborated with the teaching team to develop consistent grading criteria and resolve technically challenging assignment questions.
- Provided individualized feedback to help students identify errors in algorithmic reasoning and proof construction.

Honors

- Master of Science Award of Excellence (awarded to fewer than 5% of students), Columbia Engineering
- John Wu & Jane Sun Sunshine Scholarship, UM–SJTU Joint Institute
- Undergraduate Merit-Based Scholarship, Shanghai Jiao Tong University
- Dean's List, UM–SJTU Joint Institute

Technical Skills

Programming: Python, MATLAB, C/C++, CUDA, Julia, R
Machine Learning: TensorFlow
Hardware and Circuits: Verilog, Cadence Spectre, PSpice